



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 · STANDARD 6.G.1

Area of Trapezoids

Lesson 5-2 · Teacher Copy

Name: Type your name

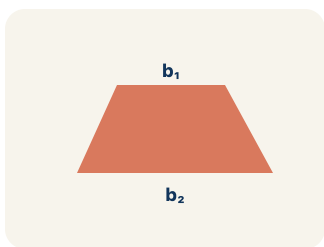
Date: Today's date

Class: Class period

Key Vocabulary

Level 2 Standard

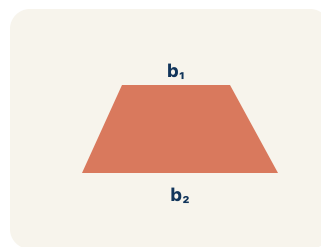
Picture first, then the word, then a plain-language meaning. Say each word out loud.



A table top wider than the bottom shelf — the top and bottom edges are parallel, the two sides slant inward

Trapezoid

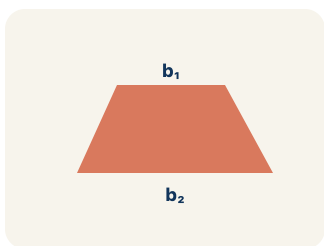
Write the definition:



If the bottom of the trapezoid is 10 ft, then $b_1 = 10$ ft in the formula $A = \frac{1}{2}(b_1 + b_2) \times h$

Base 1 (b_1)

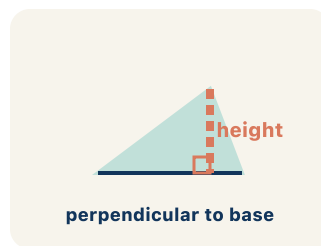
Write the definition:



If the top of the trapezoid is 6 ft, then $b_2 = 6$ ft; both bases are parallel to each other

Base 2 (b_2)

Write the definition:



A dashed vertical line from the top base straight down to the bottom base at a 90° angle — NOT the slanted side

Height

Write the definition:



Area = 15 square units

*A trapezoid with $b_1 = 10$, $b_2 = 6$, $h = 4$ has area =
 $\frac{1}{2}(10 + 6) \times 4 = 32$ sq units*

Area

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the area of a trapezoid using the formula $A = \frac{1}{2}(b_1 + b_2) \times h$.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Trapezoid

Base 1 (b_1)

Base 2 (b_2)

Height

Area



Tap any word to see what it means and a picture.

1

A quadrilateral with exactly one pair of parallel sides is a

2

One of the two parallel sides of a trapezoid is .

3

The other parallel side of a trapezoid is .

4

The perpendicular distance between the two parallel bases is the

5

The amount of flat space inside the trapezoid is its .



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?

- A. 32 sq cm
- B. 64 sq cm
- C. 24 sq cm
- D. 40 sq cm

Step 1 $A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (6 + 10) \times 4 = \frac{1}{2} \times 16 \times 4 = 32 \text{ sq cm}.$

 **Answer:** A. 32 sq cm

 We do — together

Solve this one with your class using the same steps.

Problem: A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?

- A. 5 ft
- B. 9 ft
- C. 3 ft
- D. 10 ft

Step 1 _____

Step 2 _____

Answer: _____

**You do — your turn**

Now try one on your own. Show every step.

Problem: What is the first step when finding the area of a trapezoid?

- A. Add the two bases together
- B. Multiply the two bases
- C. Subtract the shorter base from the longer base
- D. Divide the height by 2

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Blueprint 2: A trapezoidal skylight has bases of 9 ft and 7 ft and a height of 6 ft. What is its area?

- A. 48 sq ft
- B. 96 sq ft
- C. 63 sq ft
- D. 44 sq ft

Show your work:

2. Level 2 Extension: Two copies of a trapezoidal window (bases 5 ft and 11 ft) are flipped and joined to make a parallelogram with height 6 ft. The parallelogram's base is $5 + 11 = 16$ ft, so its area is 96 sq ft. What is the area of ONE trapezoid window?

- A. 48 sq ft
- B. 96 sq ft
- C. 16 sq ft
- D. 192 sq ft

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A trapezoid and a parallelogram both have a height of 8 ft. The trapezoid has bases of 5 ft and 11 ft. The parallelogram has a base of 8 ft. Which shape has the greater area? Explain your reasoning using the formulas.

Sentence starter: The trapezoid's area is ____ because $\frac{1}{2} \times (\text{ } + \text{ }) \times \text{ } = \text{ }$. The parallelogram's area is ____ because $\text{ } \times \text{ } = \text{ }$. The ____ has a greater area by ____ sq ft.

Show your work:

Reflect — Exit Ticket

A trapezoid has bases of 9 inches and 5 inches, and a height of 6 inches. What is its area?

- A. 42 sq in
- B. 84 sq in
- C. 27 sq in
- D. 42 in

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Trapezoid
2. **Notes 2:** Base 1 (b_1)
3. **Notes 3:** Base 2 (b_2)
4. **Notes 4:** Height
5. **Notes 5:** Area
6. **We Try (notes):** A. 5 ft — $A = \frac{1}{2} \times (b_1 + b_2) \times h \rightarrow 45 = \frac{1}{2} \times 18 \times h \rightarrow 45 = 9h \rightarrow h = 5 \text{ ft.}$
7. **You Try (notes):** A. Add the two bases together — *The formula is $A = \frac{1}{2} \times (b_1 + b_2) \times h$. The first step is to add the two parallel bases.*
8. **Try It 1:** A. 48 sq ft — $A = \frac{1}{2} \times (9 + 7) \times 6 = \frac{1}{2} \times 16 \times 6 = \frac{1}{2} \times 96 = 48 \text{ sq ft.}$
9. **Try It 2:** A. 48 sq ft — *One trapezoid is half the parallelogram: $96 \div 2 = 48 \text{ sq ft.}$ This matches $A = \frac{1}{2} \times (5 + 11) \times 6 = \frac{1}{2} \times 16 \times 6 = 48 \text{ sq ft}$ — the $\frac{1}{2}$ comes from the trapezoid being half of the parallelogram.*
10. **Exit Ticket:** A. 42 sq in — $A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (9 + 5) \times 6 = \frac{1}{2} \times 14 \times 6 = 42 \text{ square inches.}$